

Latent Bridge Matching: IDT-Calibrated Latent Transport for Domain-Level Artistic Transfer

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Abstract

Domain-level artistic transfer has a blunt evaluation failure: when the source is already art, doing nothing can score as target style. On the CLIP-separated WikiArt Distinct5 stress split, the reproduced SaMAM run remains below the identical-image transfer-only CLIP-S floor through its last trusted 2250-step checkpoint. We introduce *Latent Bridge Matching* (LBM), a 3.9M-parameter latent-transport model that selects target-domain endpoints during training, executes a style-conditioned vector field, and applies terminal distribution matching in VAE latent space. On Distinct5-

512, the reviewed LBM-F/K operating points exceed IDT by $+0.0244/+0.0312$ transfer-only CLIP-S at the first reproduced RTX 3060 checkpoint (1.2 minutes), whereas SaMST exceeds the same floor only in a much higher-displacement, higher-targetwise-ArtFID region. On the historical strict-750 reference surface, LBM reaches 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image, motivating a simple rule for separated art-to-art stress splits of this type: measure the unchanged-image floor before treating raw target-style affinity as stylization.

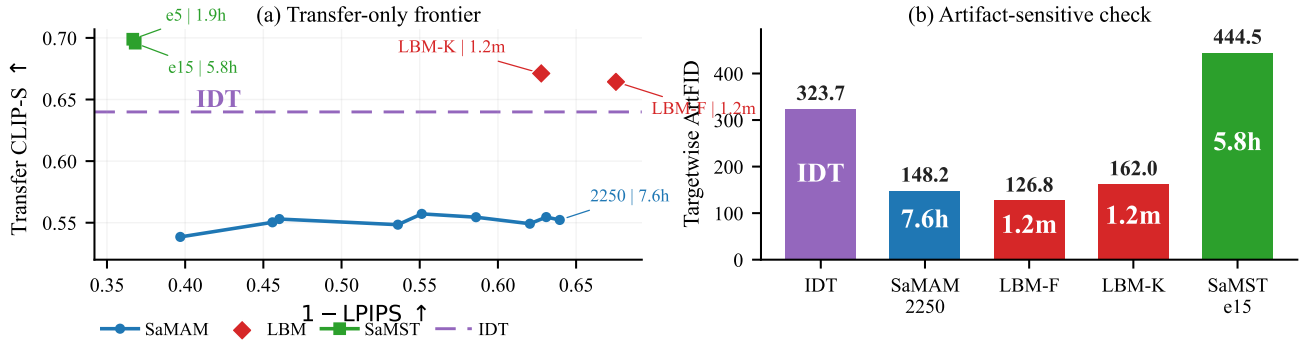


Figure 1: IDT-calibrated Distinct5-512 transfer-only operating points. Among the reproduced comparison points shown here, SaMAM stays below IDT, SaMST exceeds IDT only in a high-displacement, high-targetwise-ArtFID region, and LBM-F/K exceed IDT in a lower-damage region.

Introduction

Domain-level artistic transfer asks for a stylized image from a source image and a target style identifier. This is a compact deployment setting: the model need not receive a reference style image or run per-image optimization at inference. It also exposes a failure in common evaluation practice. When both source and target domains are artworks, an unchanged source can already obtain a strong target-style score. A method can therefore appear competitive by preserving an art-domain prior, or by making visible but misdirected edits, without moving toward the requested target domain. The first question should be signed target-style movement over the unchanged-image baseline, not raw target-style affinity.

This failure is not the familiar style/content trade-off. Classical neural style transfer introduced feature reconstruction and Gram statistics (Gatys, Ecker, and Bethge 2016);

feed-forward and attention-based arbitrary style transfer improved efficiency and local correspondence (Huang and Belongie 2017; Park and Lee 2019; Deng et al. 2022; Zheng et al. 2024). Recent systems now span reference-guided stylization (Hong et al. 2023; Huang et al. 2024; Zhang et al. 2025; Shang et al. 2025), compact multi-style models selected by style identity (Liu et al. 2024), and training-free or diffusion-prior methods (Gim, Yoo, and Uh 2024; Chung, Hyun, and Heo 2024; Deng et al. 2024; Jiang et al. 2025; Xu et al. 2025). Across these regimes, automatic metrics can conflate three different events: preserving a source that is already artistic, producing plausible art-domain texture, and moving toward the requested target style. In the style-id setting, the distinction is decisive because inference receives only a domain label.

The representation problem is geometric as well as statistical. VAE latents are compressed feature coordinates with

nontrivial local curvature, not RGB pixels. Treating target style as pointwise Euclidean endpoint reconstruction can encourage mean solutions, suppress semantically valid local rearrangements, and blur the line between preserving content and doing nothing. We instead model stylization as controlled latent transport: target-domain examples select where supervision should pull, a style-conditioned vector field decides how to move, and terminal distribution matching checks whether the integrated endpoint has target-style patch statistics.

We propose *Latent Bridge Matching* (LBM), a compact latent-transport model built around that evaluation requirement. LBM combines (i) entropic OT couplings that construct unpaired latent endpoint pairs; (ii) a style-conditioned latent vector field learned by conditional flow matching; and (iii) SA-SWD, an adaptive-projection terminal regularizer that pulls the Euler-integrated endpoint toward target-style patch statistics. A learnable style spatial prior and semantic cross-attention provide domain-level conditioning without per-image optimization or diffusion sampling. The paper also establishes a reporting rule for separated art-to-art stress splits of this type: evaluate style gain against the identical-image floor, not raw target-style affinity alone.

Our contributions are:

- **We introduce LBM, a compact latent-transport design for domain-level artistic transfer**, combining entropic OT couplings, conditional vector-field execution, and terminal distribution matching as an alternative to direct latent endpoint reconstruction.
- **We expose the Identical-Image Transfer (IDT) floor as a necessary diagnostic on separated art-to-art splits**, showing on Distinct5-512 that a reproduced baseline can visibly edit images yet score below IDT in transfer-only CLIP-S, and advocating Δ_{idt} as the companion metric that measures genuine style movement beyond that floor.
- **We report compact operating-point evidence with explicit cost**: LBM clears the Distinct5-512 IDT floor after 1.2 minutes of checkpoint-training wall time in the reproduced setup, and reaches 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image on the historical strict-750 surface. Matched controls tie weaker kinetic regularization to larger executed motion and a worse style/content trade-off.

Related Work

Classical and arbitrary neural style transfer. Neural style transfer began with optimization-based feature reconstruction and Gram-matrix style losses (Gatys, Ecker, and Bethge 2016). AdaIN later enabled real-time arbitrary transfer by aligning channel-wise feature statistics (Huang and Belongie 2017). Attention-based methods such as SANet and StyTr2 improved local style matching and long-range interaction through learned content-style correspondence (Park and Lee 2019; Deng et al. 2022). More recent reference-guided arbitrary stylization continues to refine the injection path itself: Puff-Net emphasizes

cleaner content/style feature separation, HSI replaces pointwise attention with a holistic injector, and SCSA focuses on semantic-region consistency in arbitrary semantic style transfer (Zheng et al. 2024; Zhang et al. 2025; Shang et al. 2025). These works mainly ask how a style exemplar should be fused at inference time. Our problem is different: the main protocol in this paper does not provide a target style image at inference, so the central question is how a domain-level style control should be executed without exemplar-conditioned correspondence.

Efficient multi-style transfer and compact style representations. Multi-style transfer amortizes many styles into one compact model rather than solving a fresh arbitrary-style problem for every target image. SaMST is the closest recent representative: it separates style modeling from style transfer through pluggable style representations and a style-aware transfer network, explicitly targeting the quality-efficiency trade-off for many-style deployment (Liu et al. 2024). This line is the most relevant tokenizer-side baseline for our work. Throughout this paper, “style tokenizer” means a style-id-conditioned control map rather than an image tokenizer, a target-image encoder, or a diffusion-token controller. Our contribution is not another pluggable codebook alone. Instead, we ask whether a compact style representation remains executable after it passes through a latent transport field, so that style-side distinctions survive downstream rendering strongly enough to produce real style movement beyond the unchanged-image baseline.

State-space and Mamba-based style transfer. Recent work also revisits style transfer through state-space backbones. Mamba-ST and SaMam argue that linear-complexity state-space operators can recover global receptive fields more efficiently than quadratic attention, improving the efficiency-quality frontier of arbitrary or many-style transfer (Botti et al. 2025; Liu et al. 2025). These papers are important because they shift the architectural question from “CNN or transformer” to “which global-mixing primitive is most efficient.” Our comparisons to SaMam take this line seriously. The distinction is that LBM does not claim a new global-mixing backbone; its main contribution is objective-side: OT-coupled endpoint construction, transport-field learning, and semantic-aligned terminal distribution matching.

Training-free and diffusion methods. Training-free and diffusion-based approaches provide a different route to flexible style transfer. CAST uses VQ autoencoders for content-adaptive training-free stylization (Gim, Yoo, and Uh 2024). StyleID and Z* show that large diffusion priors can be adapted for artistic style transfer without retraining by manipulating attention features or attention weights at inference time (Chung, Hyun, and Heo 2024; Deng et al. 2024). SMS further pushes this direction by casting stylization as diffusion-side style distribution matching with additional spectrum-aware and semantic-aware refinement (Jiang et al. 2025). StyleSSP sharpens the same training-free diffusion line by improving the sampling start point to reduce layout drift and style-image content leakage (Xu et al. 2025).

These methods can be visually strong, but they generally assume a style reference image and often inherit the inference cost or calibration issues of large generative priors. Our work instead targets compact retrainable latent integration with style-id conditioning rather than iterative diffusion sampling, prompt engineering, or per-style inference-time optimization.

Distribution matching and perceptual evaluation. Our objective is motivated by optimal transport and distribution matching. Sinkhorn distances and computational optimal transport provide efficient tools for approximate transport problems (Cuturi 2013; Peyré and Cuturi 2019), while the Schrodinger problem connects entropic transport to stochastic path measures (Léonard 2014). We use SWD as a lightweight terminal distribution objective and instantiate it in the current mainline with adaptive projection routing through SA-SWD. OT supplies the mini-batch endpoint couplings in latent space, and the formal analysis below grounds why this combination yields smooth supervision, bounded path energy, and stable few-step integration in the operating regime studied here. Evaluation itself remains unsettled in arbitrary style transfer: prior studies report inconsistent protocols across papers and show that commonly used automatic metrics are style-dependent and require calibration to human preference (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our metric claim is therefore focused rather than universal. For separated art-to-art stress splits of this type, we make the unchanged-image prior explicit through an ‘idt’ reference and report Δ_{idt} so that target-style movement can be separated from preserved art-domain priors. In the Distinct5 results, the paper-facing ArtFID column is targetwise ArtFID; aggregate ArtFID is retained only as an auxiliary art-domain/identity diagnostic outside the main comparison table. We therefore combine perceptual and distributional measures including LPIPS (Zhang et al. 2018), FID (Heusel et al. 2017), KID (Bińkowski et al. 2018), DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), MANIQA (Yang et al. 2022), and ArtFID (Wright and Ommer 2022). This broader metric set is important because raw CLIP-style and structural scores may fail to penalize grain-like artifacts or may partially reflect art-domain priors already present before stylization.

Method

Overview

LBM is a conditional-flow latent transport model with OT-coupled endpoints in VAE space for domain-level artistic transfer. Let $z_0 \in \mathbb{R}^{4 \times 32 \times 32}$ denote the VAE latent of a content image and let s denote a target style domain identifier. It comprises four coupled components: (i) entropic OT constructs unpaired content and target-style endpoints within each minibatch; (ii) a time-conditioned transport field $v_\theta(z_t, t, s)$ realizes the corresponding style-conditioned transport; (iii) kinetic regularization discourages unnecessary latent displacement; and (iv) SA-SWD supplies terminal distribution matching at the integrated endpoint. At inference, the learned vector field is integrated with a fixed-step solver and decoded by the VAE decoder. In this de-

composition, the tokenizer specifies the target region of style space, and the backbone is the executor that realizes the corresponding transport field.

In the reproduced Distinct5 setting, endpoint-conditioned transport supervision is instantiated as OMF endpoint-delta training with terminal matching and kinetic regularization. We therefore analyze LBM at the transport-design level and name objective-family differences only when they matter empirically.

Unpaired latent-domain sampling and OT coupling

We train directly on precomputed VAE latents without paired supervision. Each minibatch samples content latents $\{z_c\}$ from all source domains and target-style latents $\{z_s\}$ from the requested target domain. Identity pairs are naturally included through uniform target-domain sampling, which stabilizes the learned transport around same-domain mappings.

Rather than using random target assignments, we construct a mini-batch transport coupling using the sliced-Wasserstein cost. For a batch of content latents $\{z_c^{(i)}\}$ and target-style latents $\{z_s^{(j)}\}$, the SWD cost matrix is

$$C_{ij} = \text{SWD}(z_c^{(i)}, z_s^{(j)}). \quad (1)$$

An entropic optimal-transport plan is then obtained via the Sinkhorn algorithm:

$$\pi^* = \arg \min_{\pi} \sum_{ij} C_{ij} \pi_{ij} + \epsilon H(\pi), \quad (2)$$

where $H(\pi)$ is the entropy regularizer. Each content latent is matched to its coupled target $\tilde{z}_1 \sim \pi^*(z_s | z_c)$. This coupling provides style-domain endpoints without requiring spatially aligned targets.

Time-conditioned latent transport supervision

The core of the method is a time-conditioned velocity field $v_\theta(z_t, t, s)$ that transports latents between OT-coupled endpoints along a stochastic interpolation path. At a randomly sampled time $t \sim \mathcal{U}(0, 1)$, the interpolated state is

$$z_t = (1 - t)z_0 + t\tilde{z}_1 + \sigma\sqrt{t(1 - t)}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

and the target velocity is the derivative of the transport:

$$u_t = \tilde{z}_1 - z_0 + \sigma \frac{1 - 2t}{2\sqrt{t(1 - t)}} \epsilon. \quad (4)$$

The flow-matching objective supervises the network to predict this velocity:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{z_0, \tilde{z}_1, t, \epsilon} \rho(v_\theta(z_t, t, s), u_t). \quad (5)$$

Here ρ is the pointwise velocity penalty. The implementation supports MSE, Huber, and L_1 variants in the direct velocity-regression path. Equation (5) defines the parent transport objective; in the reproduced Distinct5 setting, training instantiates that objective through OMF-style endpoint-delta supervision together with terminal matching and kinetic regularization. The current evidence favors this OT + W_1 -style mainline over isolated endpoint-only pointwise supervision.

Latent Bridge Matching

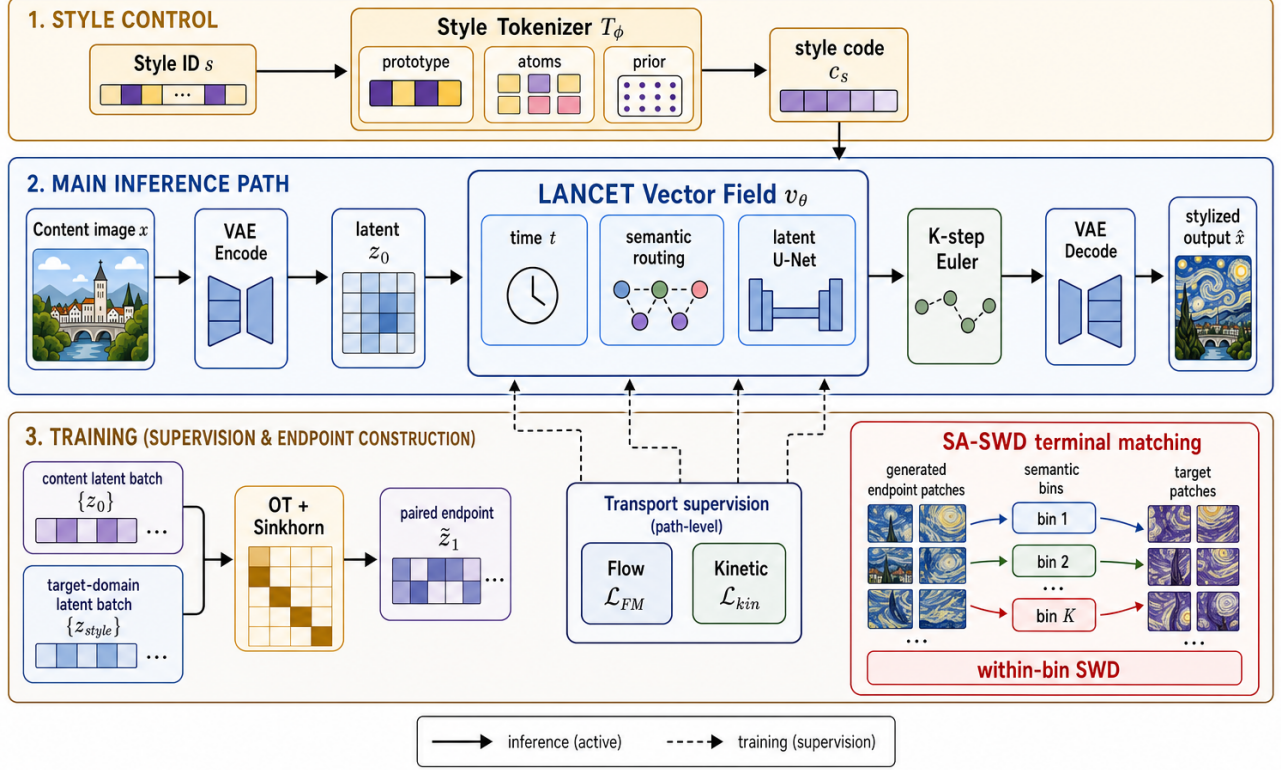


Figure 2: Overview of Latent Bridge Matching. The top band shows style-side control, where the style tokenizer maps the target style identifier into a compact style code. The middle band is the active inference path: content is encoded into a latent, passed through the LANCET vector field with time conditioning and semantic routing, integrated by K -step Euler updates, and decoded into the stylized output. The bottom band shows training-only constraints: target-domain latents enter through mini-batch OT endpoint construction and OMF-style endpoint/field supervision, while the SA-SWD inset makes explicit how semantic bins define terminal within-bin SWD matching. Target-domain latents are therefore training-side supervision only, not inference-time conditioning inputs.

Architecture. The time-conditioned backbone uses sinusoidal time embeddings added to the style code. The implementation is a U-Net-like latent trunk with four semantic cross-attention body blocks at 16×16 resolution. The target style is represented by a learnable spatial prior $P_s \in \mathbb{R}^{128 \times 16 \times 16}$. Semantic cross-attention injects the style prior into content features, while skip connections preserve spatial structure.

Style tokenizer as executable control

The style-side interface is deliberately separated from the latent renderer. In the main domain-level protocol, the tokenizer receives only a target style identifier and learned style parameters available at inference:

$$c_s = T_\phi(s), \quad v_\theta = F_\theta(z_t, t, c_s). \quad (6)$$

It does not read a target image or target latent from the current evaluation batch. Target-domain latents remain on the loss side of the graph through OT coupling and SA-SWD,

but they are not conditioning evidence for T_ϕ . This boundary is important: otherwise training would solve a reference-guided problem while evaluation reports a style-id problem. In this sense, T_ϕ names where transport should go, while F_θ decides how much latent motion is actually needed to get there. Throughout this paper, “style tokenizer” means a style-id-conditioned control map, not an image-token or diffusion tokenizer and not a target-image encoder.

The simplest tokenizer is a per-style code table. Our current implementation generalizes this interface without changing the LANCET consumer:

$$c_s = p_s + \gamma \sum_{k=1}^K \alpha_{s,k} a_k, \quad \alpha_s = \text{softmax}(\ell_s / \tau), \quad (7)$$

where p_s is an optional direct prototype, $\{a_k\}$ are shared style atoms, and γ controls the residual atom contribution. In the current mainline, this interface is paired with a prototype-aware latent target queue so that the renderer

receives a stable style carrier without changing its single-code input contract. The tokenizer therefore exposes an executable style vocabulary while preserving the compact interface required by the existing renderer.

For execution, the backbone may also use style spatial priors and content-conditioned routing. These modules modulate where a target-style control is applied, not what target style is requested. This distinction matches the empirical diagnosis: the key question is not raw tokenizer capacity in isolation, but whether the style carrier remains executable after it passes through a content-conditioned renderer. Prototype-aware target queues and content-guided routing improve that executed style-content frontier, which is why we phrase tokenizer evidence at the level of *executable representation*: code geometry matters only insofar as generated latent residuals remain distinguishable after LANCET consumes the code.

Complementary roles. The components address distinct subproblems. OT coupling provides target-style endpoints for unpaired supervision, but not transport dynamics. Transport supervision learns how to move toward those endpoints, but by itself does not guarantee that the integrated endpoint matches target-domain texture statistics. SA-SWD regularizes this endpoint mismatch, while semantic cross-attention provides local routing for where style is written and supplies a principled heuristic for projection selection. The destructive ablations directly verify the roles of terminal distribution matching and kinetic regularization. On the current Distinct5 packet, the dominant gain comes from distributional terminal matching itself, which avoids pointwise latent collapse, while semantic routing refines how those projection axes are chosen.

Adaptive-projection terminal distribution matching

In addition to flow-matching supervision, a terminal loss regularizes the Euler-integrated endpoint toward the target style distribution. Let $\Phi_\theta(z_0, s)$ denote the endpoint obtained by integrating v_θ from z_0 with style s . The terminal loss is

$$\mathcal{L}_{\text{term}} = \text{SWD}_1(\Phi_\theta(z_0, s), \tilde{z}_1), \quad (8)$$

where the SWD compares generated endpoint patches with coupled target patches. We use semantic cues to choose SA-SWD projection directions. Let K_s denote the keys produced by semantic cross-attention for target style s . We rank spatial target positions by the average magnitude of K_s and use the corresponding target latent patch vectors as normalized projection directions. This defines the SA-SWD instantiation used in the current mainline: terminal distribution matching is still computed by sorted one-dimensional projections, but the projection axes are adaptively chosen from the same routing signal that injects style into the backbone.

The complete training objective combines transport supervision, terminal distribution matching, and kinetic path regularization:

$$\mathcal{L} = \mathcal{L}_{\text{transport}} + \lambda_{\text{term}}\mathcal{L}_{\text{term}} + \lambda_{\text{kin}}\mathcal{L}_{\text{kin}}, \quad (9)$$

where $\mathcal{L}_{\text{transport}}$ is instantiated as endpoint-conditioned flow supervision: Eq. (5) for the historical bridge lineage and the resolved OMF-style endpoint-delta term for the current Distinct5 mainline. The kinetic loss $\mathcal{L}_{\text{kin}} = \mathbb{E}\|v_\theta\|_2^2$ penalizes excessive latent displacement. The primary configuration uses $\lambda_{\text{term}} = 20.0$ and $\lambda_{\text{kin}} = 1.0$. Color, cycle, NCE, and repulsive losses are disabled in the main setting. This three-part loss separates the forces that drive style from those that preserve content.

Metric choice in latent space. The training objective does not treat all latent discrepancies as Euclidean reconstruction errors. The kinetic term uses an L_2 penalty on velocity because it represents path energy, while terminal endpoint matching uses Wasserstein-1 style discrepancies over sorted projections. This separation is deliberate in compressed VAE latents: pointwise endpoint MSE would reward mean solutions and over-penalize semantically valid local rearrangements, whereas SA-SWD preserves distributional supervision without requiring paired spatial correspondence. The current reproduced Distinct5 checkpoints therefore place style pressure on OT endpoint construction plus W_1 -style terminal matching rather than on latent endpoint MSE, which is precisely the design choice that avoids mean-seeking endpoint regression in the measured setting.

Theoretical grounding of latent transport. Section establishes three properties of the model: Theorem 1 connects the training kinetic loss to the inference-time path action, Theorem 2 bounds the Euler discretization error at $O(1/K)$, and Theorem 3 provides a Lipschitz-continuity rationale for OT-based endpoint coupling.

More concretely, let $\mathcal{P}_r(\hat{z}_1)$ be the set of latent patches of size r extracted from the generated endpoint, and let $\mathcal{P}_r(z_s)$ be the corresponding patch set from target-style samples. For projection directions $\{\theta_m\}_{m=1}^M$, the empirical terminal objective is

$$\mathcal{L}_{\text{swd}} = \sum_{r \in \mathcal{R}} \frac{1}{M} \sum_{m=1}^M \left\| \text{sort}(\mathcal{P}_r(\hat{z}_1)\theta_m) - \text{sort}(\mathcal{P}_r(z_s)\theta_m) \right\|_1. \quad (10)$$

Here $\mathcal{R} = \{3, 5, 7, 15\}$ in the primary configuration. The loss therefore asks whether the generated endpoint has target-style patch statistics along semantic routing directions, not whether it matches any particular artwork spatially.

Formal analysis

We provide three formal results that characterize key properties of the proposed method. Full proofs appear in the supplement. For brevity we state the results and their empirical support.

Notation. Recall the OMF training regime: $\hat{v} = v_\theta(z_0, 1, s)$, $\hat{z}_1 = z_0 + \hat{v}$. At inference, integration follows $dz/dt = v_\theta(z, t, s)$ with K -step Euler: $z_{k+1} = z_k + \Delta t v_\theta(z_k, t_k, s)$ where $\Delta t = 1/K$, $t_k = (k + 0.5)\Delta t$. Define the path action $\mathcal{A}(v) = \int_0^1 \mathbb{E}\|v_\theta(z(t), t, s)\|^2 dt$ and the training kinetic loss $\mathcal{L}_{\text{kin}} = \mathbb{E}\|v_\theta(z_0, 1, s)\|^2$.

Design interpretation. Under bounded temporal drift and Lipschitz continuity, endpoint kinetic regularization acts as a tractable proxy for inference-time path action. Theorem 1 formalizes why a lightweight endpoint penalty is sufficient to straighten trajectories in the operating regime studied here, without requiring expensive multi-step supervision.

Theorem 1 (Kinetic regularization as a path-energy proxy). Assume that for a fixed content z_0 the velocity norm varies at most δ with t , i.e., $\sup_t \|v_\theta(z_0, t, s)\| - \|v_\theta(z_0, 1, s)\| \leq \delta$, and that v_θ is L -Lipschitz in z . Let M bound $\|v_\theta\|$ along the trajectory. Then the path action decomposes as

$$\mathcal{A}(v) = \mathcal{L}_{\text{kin}} + \varepsilon_t + \varepsilon_z, \quad (11)$$

where $|\varepsilon_t| \leq 2\delta\sqrt{\mathcal{L}_{\text{kin}}} + \delta^2$ and $|\varepsilon_z| \leq LM$. If δ and LM are small, the training kinetic loss provides a tight proxy for the inference-time path energy.

Empirical support. Experimentally (128-step integration, D0 model): $\mathcal{L}_{\text{kin}} \approx 1017$, $\mathcal{A}(v) \approx 698$, yielding $|\varepsilon_t + \varepsilon_z| \approx 319$. Removing $\mathcal{L}_{\text{kin}}$ (D2) reduces the path length ratio from 0.98 to 0.94, confirming the kinetic term controls path regularity.

Theorem 2 (Euler discretization bound). Under the L -Lipschitz condition on v_θ , the K -step Euler integration error satisfies

$$\|z(1) - z^{(K)}\| \leq \frac{C}{K}, \quad C = \frac{1}{2}(ML_t + M^2L) \frac{e^L - 1}{L}, \quad (12)$$

where L_t bounds the t -derivative of v_θ .

Empirical support. Measured over 160 content latents (D0 model, 256-step reference): $\|z^{(K)} - z^{(256)}\| = \{10.7, 3.80, 1.74, 0.84, 0.40, 0.19, 0.084, 0.027\}$ for $K = 1, 2, 4, 8, 16, 32, 64, 128$; the error halves when K doubles, confirming the $O(1/K)$ scaling. Because VAE latents exhibit non-trivial local curvature, few-step solvers can otherwise accumulate visible drift. Here the measured $O(1/K)$ decay shows that the default few-step Euler regime remains strictly controlled in practice. At $K = 12$ (the default), the error is < 0.6 —negligible relative to the style-transfer effect.

Theorem 3 (OT assignment continuity). Let the SWD cost $c(\cdot, \cdot)$ be L_c -Lipschitz and let π^* be the ε -Sinkhorn plan on a batch of size B . Let D denote the diameter of the target latent set. Then, for any two content latents z_0, z'_0 , the expected matched targets satisfy

$$\|\mathbb{E}_{\pi^*}[\tilde{z}_1 | z_0] - \mathbb{E}_{\pi^*}[\tilde{z}_1 | z'_0]\| \leq \frac{D \cdot L_c}{\varepsilon} \|z_0 - z'_0\| + O(\varepsilon). \quad (13)$$

Thus the OT assignment map is (DL_c/ε) -Lipschitz: nearby content latents receive nearby matched targets. Random uniform coupling provides no such guarantee. The Lipschitz property implies smoother supervision gradients and more directionally coherent displacement vectors across the batch.

Empirical support. Compared over 640 content latents: OT coupling yields mean cosine similarity to the batch-mean direction of 0.150, vs. 0.124 for random coupling, a 21% improvement. The transport cost is also reduced by 4.2% (0.119 vs. 0.124).

Experiments

Protocol hierarchy and metrics

We organize the evidence into two non-interchangeable result families, with Distinct5-512 as the primary stress benchmark and the historical strict-750 packet as contextual support. This separation is necessary because the same scalar pair, CLIP-style and LPIPS, can be misleading when datasets, image resolution, or baseline training protocols change.

Historical strict-750. The main legacy protocol uses five domains—photo, Hayao, Monet, Van Gogh, and Cezanne—and evaluates all 5×5 source-target directions with 30 source images per source domain, for 750 generated outputs. This is the protocol used for comparisons to SaMST, S2WAT, StyleID, AdaIN, StyTr2, CAST, AesFA, and AesPA-Net.

Distinct5-512 stress split. The new stress split uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. These five classes were chosen as the most strongly separated subset under a CLIP-style screening pass over WikiArt categories, rather than being hand-picked for LBM. Evaluation again uses all 5×5 directions over 750 generated images. This split complements the historical table with a stronger domain-separation test and an explicit identical-image baseline.

We report CLIP-style (CLIP-S) and CLIP-content (CLIP-C) using CLIP ViT-B/32 embeddings (Radford et al. 2021), together with LPIPS-content and

$$\text{EC} = \text{CLIP-style} \cdot (1 - \text{LPIPS}). \quad (14)$$

For Distinct5-512 we additionally report style gain over the unchanged-image reference,

$$\Delta_{\text{idt}} = \text{CLIP-S}(\text{method}) - \text{CLIP-S}(\text{idt}), \quad (15)$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. We use this quantity to define the metric-stress regime exposed by Distinct5-512: raw CLIP-style should not be treated as sufficient evidence of successful target-style transfer if a method either fails to exceed the identical-image prior ($\Delta_{\text{idt}} \leq 0$) or exceeds it only by moving into a clearly damaged LPIPS region while also deteriorating broad artifact diagnostics. In this paper, CLIP-S is interpreted as target-style affinity, LPIPS as content displacement, and targetwise ArtFID as a target-domain plausibility and artifact diagnostic rather than a direct style-gain score. Aggregate ArtFID is kept only as an auxiliary art-domain/identity diagnostic outside this main Distinct5 comparison. In particular, lower targetwise ArtFID alone does not certify useful stylization on separated art-to-art transfer: its content term rewards source preservation, and its distributional term can reward remaining in a plausible target-domain art manifold even when the requested target-style movement is still below the unchanged-image baseline. EC is used only as a Pareto summary; claims are checked against CLIP-S and LPIPS individually. For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, shallow Gram micro

statistics, and KID where available. All strict-750 baselines use the same source images, target domains, preprocessing, and identity directions. For arbitrary-style methods, one target-domain reference image is supplied per target style; for style-id LBM, the model receives only the target domain identifier. The ‘idt’ control is introduced here as a split-specific diagnostic baseline for art-to-art transfer: our claim is that Distinct5 makes explicit the role of the unchanged-image prior on this split, not that any single existing metric should be discarded.

Distinct5-512 stress benchmark

Distinct5-512 is a stress benchmark for separated art-to-art transfer. All five domains come from standard WikiArt classes, and they are selected only because they are among the most separated under a CLIP-style screening pass. In this regime, unchanged output is no longer a plausible success proxy: the *idt* baseline already reaches CLIP-S 0.6801 while doing no transfer at all. Signed style gain is therefore the primary sanity check. Under the shared full 5×5 / 750-output protocol, LBM defines the strongest positive- Δ_{idt} CLIP-S / LPIPS frontier among the reproduced compact domain-level operating points. The content-preserving LBM-F checkpoint reaches CLIP-S 0.6969 and LPIPS 0.3186 after 1.2 minutes of checkpoint training, while LBM-K raises style to 0.7010 at LPIPS 0.3623 and remains well below the high-damage regime. SaMST-512 pushes raw style higher to 0.7247, but only by entering that regime directly, with LPIPS 0.6255 and targetwise ArtFID 395.7; the retained e5/e15 packet is already near a plateau on transfer CLIP-S and LPIPS, with less than 0.004/0.002 difference between the checkpoints. At the trusted SaMAM 2250 checkpoint, transfer CLIP-S remains below the idt floor even though targetwise ArtFID improves to 146.1 on the full scope and 148.2 on the transfer-only scope. This is the evaluation failure exposed by Distinct5: lower targetwise ArtFID can still coincide with zero or negative target-style gain when a method preserves source structure and stays inside a plausible target-art manifold without moving toward the requested target style. Distinct5 therefore turns the paper’s evaluation claim into an explicit operating-point test: target-style movement should be demonstrated relative to idt, not inferred from absolute CLIP-S or ArtFID alone.

Table 2 makes the frontier interpretation concrete on representative transfer-only pairs from the reviewed packet: SaMST can produce locally strong target-style jumps, but those gains are not stable across directions and often require much larger displacement than the balanced LBM-F point, while SaMAM still often edits without moving toward the requested target style.

Historical strict-750 comparison

We then place the historical strict-750 packet as contextual support for the older comparison surface. Table 3 establishes LBM as the strongest compact content-preserving operating point among the reproduced baselines in this historical strict-750 comparison. StyleID reaches the highest raw CLIP-style but collapses content. SaMST is marginally higher on CLIP-S and CLIP-C, yet LBM

Table 1: Distinct5-512 all-pairs stress benchmark. All rows use the same 5×5 evaluation over 750 generated images. ‘idt’ is the unchanged-image reference copied into every target slot. Δ_{idt} is the full-scope style gain over idt and $\Delta_{idt, tr}$ is the transfer-only off-diagonal gain. ‘tw-ArtFID’ denotes targetwise ArtFID. SaMAM is reported only at the trusted 2250-step checkpoint; later checkpoints are excluded from the active manuscript path. The final column is operating-point context only: SaMAM and SaMST report cumulative wall time to the shown checkpoint with evaluation excluded, whereas LBM reports the first reviewed checkpoint on the reproduced RTX 3060 schedule; it is not a normalized time-to-parity measure.

Method	CLIP-S $\uparrow$	LPIPS $\downarrow$	EC $\uparrow$	tw-ArtFID $\downarrow$	$\Delta_{idt} \uparrow$	$\Delta_{idt, tr} \uparrow$	Wall time to point
idt (no-op)	0.6801	0.0000	0.6801	216.5	0.0000	0.0000	0
SaMAM 2250	0.5811	0.3538	0.3755	146.1	-0.0990	-0.0877	7.6h
SaMST-512 e15	0.7247	0.6255	0.2714	395.7	+0.0446	+0.0558	5.8h
LBM-F e1	0.6969	0.3186	0.4748	122.6	+0.0168	+0.0244	1.2m
LBM-K e1	0.7010	0.3623	0.4470	157.2	+0.0209	+0.0312	1.2m

Table 2: Representative Distinct5-512 transfer-only pairs from the same reviewed packet. Cells report CLIP-S / LPIPS for the unchanged-image reference, SaMAM-2250, SaMST e15, and the balanced LBM-F e1 point. The table is meant as local sanity checking rather than a new aggregate leaderboard. Abbreviations: ER = Early Renaissance, Imp = Impressionism, Min = Minimalism, Roc = Rococo, Ukiyo = Ukiyo-e.

Pair	No-op	SaMAM-2250	SaMST e15	LBM-F e1
ER→Min	0.607	0.566 / 0.387	0.863 / 0.644	0.710 / 0.498
ER→Ukiyo	0.663	0.571 / 0.321	0.748 / 0.195	0.724 / 0.396
Imp→Min	0.602	0.623 / 0.585	0.892 / 0.731	0.823 / 0.483
Roc→Ukiyo	0.515	0.539 / 0.542	0.582 / 0.604	0.599 / 0.295

improves LPIPS and EC while also winning the artifact-sensitive metrics reported below. Relative to S2WAT and AdaIN, LBM preserves substantially more content at comparable or stronger style. Within this reproduced mixed-family strict-750 packet, LBM is the strongest compact content-preserving retrainable operating point; the claim is bounded to this table rather than intended as a same-interface ranking across all style-transfer regimes.

Cost matters here as deployment evidence, not as a normalized time-to-parity contest. The reproduced operating-point records are 310s for the reported LBM checkpoint, 6769s for the reproduced SaMST operating point, and 10600s for S2WAT. Historical strict-750 inference is 114ms/image for LBM, 53ms/image for SaMST, and 804ms/image for StyleID. The practical reading is straightforward: LBM is not the fastest feed-forward stylizer in absolute milliseconds, but it remains in a compact single-image interactive regime at 114ms/image while avoiding StyleID’s diffusion-scale inference cost and keeping a 3.91M-parameter footprint. We therefore report cost as explicit operating-point context instead of collapsing unlike training protocols into a single efficiency slogan.

Table 3: Historical strict-750 comparison in grouped family format. Higher is better for CLIP-S, CLIP-C, and EC; lower is better for LPIPS. Artifact-sensitive metrics are reported separately in Table 5.

Metric	CNN-based			Transformer-based			Training-free		Latent transport
	AesPA	AesFA	AdaIN	StyTr2	S2WAT	SaMST	CAST	StyleID	LBM
CLIP-S $\uparrow$	0.5976	0.6496	0.7130	0.6374	0.7139	0.7194	0.6652	0.7597	0.7161
CLIP-C $\uparrow$	0.5009	0.5497	0.6990	0.6001	0.7465	0.8193	0.5562	0.5519	0.8086
LPIPS $\downarrow$	0.8215	0.7392	0.6298	0.6345	0.5263	0.4664	0.7263	0.7497	0.4514
EC $\uparrow$	0.1067	0.1694	0.2639	0.2330	0.3382	0.3839	0.1821	0.1902	0.3928

Table 4: Recorded operating-point training and inference cost on the historical strict-750 protocol. All timings are wall-clock on a single RTX 4070 Laptop GPU. Training time is reported to the selected operating point used in this paper as an explicit practical-footprint record; inference reports 750-image end-to-end runtime and milliseconds per image.

Method	Train (s)	Infer-750 (s)	ms/img	Params
LBM	310	85.4	114	3.91M
SaMST	6769	39.8	53	$\sim$ 6M
S2WAT	10600	—	—	$\sim$ 7M
CAST	1760	75.5	101	7.01M
StyleID	0	603.3	804	—
AdaIN v32k	9220	9.3	12	—

Table 5: Artifact-sensitive comparison on the historical strict-750 packet among high-quality methods.

Metric	Ours e7	SaMST	S2WAT
MUSIQ $\uparrow$	49.2059	36.0950	36.5256
MANIQA $\uparrow$	0.4057	0.3139	0.1754
DISTS-content $\downarrow$	0.2477	0.2943	0.2942
HF-Patch-KID $\downarrow$	4.1694	6.7598	12.6623
FFT slope error $\downarrow$	0.5473	1.0536	0.7017
Gram micro $\downarrow$	0.0798	0.0947	—

Gram micro was not recomputed for S2WAT in the current evaluation protocol.

Artifact-sensitive diagnosis

The artifact claim is clearest on the historical 256-resolution packet, where SaMST is the closest strict-750 efficient baseline to LBM on the core metrics and also the closest recovered baseline with locally readable outputs for crop-based diagnosis. Figure 3 uses fixed matched crop boxes rather than free-floating zooms. The top grids show the full source-target matrix, while the lower row magnifies the exact boxed regions. On these matched cells, SaMST often preserves coarse layout but fills high-frequency regions with muddy grain, line-boundary contamination, and broken local brush continuity; LBM is less aggressive in raw style, but cleaner on the same spatial evidence.

Table 5 shows that LBM is not merely winning by leaving the image unchanged. Relative to SaMST, it improves MUSIQ from 36.10 to 49.21, MANIQA from 0.314 to

0.406, DISTS-content from 0.294 to 0.248, HF-Patch-KID from 6.76 to 4.17, and FFT slope error from 1.054 to 0.547. A paired bootstrap over the 750 per-image scores for Ours gives 95% intervals of [0.4467, 0.4563] for LPIPS and [0.7091, 0.7234] for CLIP-S. The photo-to-art subset ($n = 120$) yields CLIP-S 0.6716 and LPIPS 0.4882; the non-identity subset ($n = 600$) yields CLIP-S 0.6911 and LPIPS 0.4609.

Tokenizer and representation ablations

These ablations make the tokenizer conclusion sharper than a parameter-count story. Class-local prototypes and global atoms did not beat the baseline frontier, so merely enlarging the tokenizer is insufficient. Content-guided spatial routing helped, which indicates that execution location matters. The largest improvement came from a prototype-aware latent target queue, which reduces target-distribution noise before the terminal loss is applied. Finally, content-adaptive atom routing produced the best style score but regressed LPIPS, suggesting that stronger style-side separation can still demand larger executed motion. A plausible next tokenizer direction is therefore to test a two-factor interface rather than a larger undifferentiated code: a target-style carrier that remains distinguishable after execution, plus a content-risk gate that bounds how much of that carrier should be applied.

Non-training probes support the same diagnosis boundary. On the historical e8 checkpoint, tokenizer codes have near-full rank for five styles (effective rank 3.986, mean off-diagonal cosine 0.015), while generated latent residuals are much more aligned (effective rank 3.324, mean off-diagonal cosine 0.725). Residual variance is also content dominated: source-image variance explains 0.9501 of raw generated-delta variance, while target style explains only 0.0283 before source-image removal but 0.5664 after removing per-source-image residual means. On the matched Distinct5 L-e1 localization packet, executor-side refresh yields the larger reviewed improvement than style-side refresh alone, which shifts immediate bottleneck suspicion toward execution rather than raw tokenizer-code weakness on that local surface. A separate matched Distinct5 successor probe reaches the same boundary from the execution side: tokenizer-code geometry maps only moderately into executed geometry, while realized style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. Taken together, these probes show that distinguishable tokenizer codes do not automatically remain distinguishable after content-conditioned

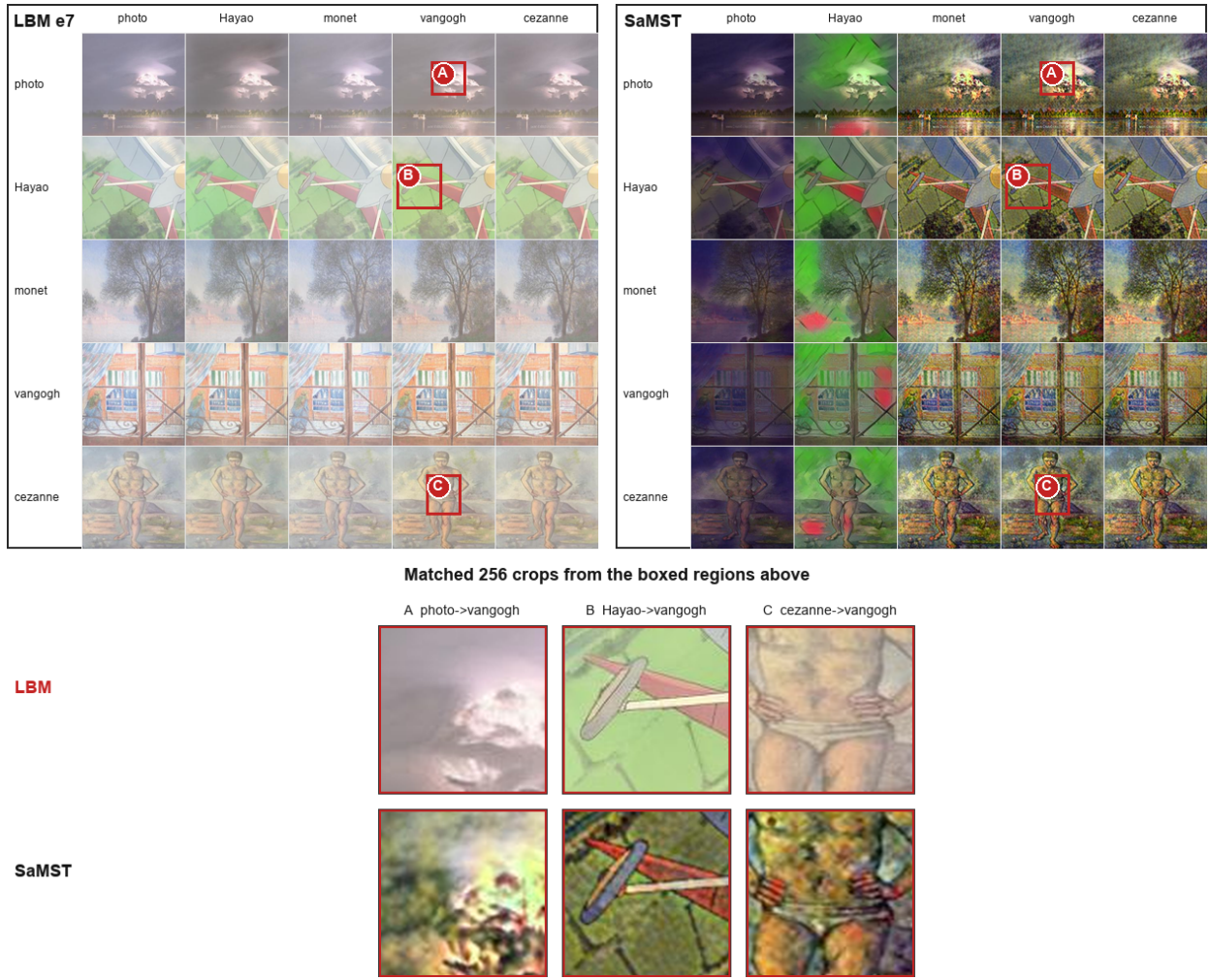


Figure 3: Matched 256-resolution artifact diagnosis on the historical strict-750 packet. Top: full source-target grids for LBM epoch 7 and SaMST with three fixed crop boxes. Bottom: magnified crops from the exact boxed regions. SaMST is shown as the closest efficient strict-750 256-resolution baseline with valid local outputs; boxes A–C highlight luminous breakup, line-boundary contamination, and body-contour corruption, respectively.

execution. They justify evaluating tokenizer quality through executed control rather than code separability alone, but they remain local to the current L-e1 / successor surfaces and do not restore H-family continuity or prove that tokenizer-side weakness is absent.

Mechanism ablations

Removing terminal distribution matching gives excellent content preservation but weak style, showing that terminal distribution alignment is a primary style driver in the current mainline. Removing kinetic regularization preserves style but collapses content, confirming that path-energy regularization is essential. Strong color loss damages the trade-off, so naive color matching is not used in the mainline. Trajectory straightness analysis gives the same reading: the full model has PLR 0.98, while removing SA-SWD produces near-linear transport to the OT target (PLR 0.9995) and removing kinetic regularization reduces straightness to PLR

0.94 with content collapse. A landed same-family Distinct5 H packet reaches the corresponding bounded path-stability read under matched epoch_0001 checkpoints: weakening kinetic regularization increases endpoint/path/peak motion from 80.24 / 80.28 / 80.41 in H_{base} to 111.71 / 111.72 / 111.83 in H_{k025} and 122.42 / 122.40 / 122.51 in H_{k000} , while CLIP-S / LPIPS shift from 0.6891 / 0.4272 to 0.6825 / 0.4600 and 0.6790 / 0.4862. We use this only as bounded same-family support that kinetic regularization acts as a practical path stabilizer in the current OMF / field regime, not as a broader theorem or cross-family generalization.

Sensitivity and efficiency

The 40-run category-weight sweep trained 20 sampling recipes with $K \in \{1, 2\}$ for eight epochs each, evaluating every epoch for 320 total checkpoints. The best EC result was K2 balanced default at epoch 3 with CLIP-S 0.6980, CLIP-C 0.8727, LPIPS 0.3777, and EC 0.4343. The

Table 6: Distinct5-512 tokenizer and representation ablations. The results show that capacity alone is not the issue; target selection, content-guided execution, and sparse routing determine whether the representation is executable.

Variant	Best epoch	CLIP-S	LPIPS	Interpretation
Direct atom residual baseline	8/1	0.6876	0.4468	weak baseline; simple residual atoms do not solve representation
Class-local prototypes	8/1	0.6849	0.4464	more per-class capacity, no better style geometry
Global VQ atoms	8	0.6873	0.4446	small LPIPS gain, no style breakthrough
Content-guided spatial routing	2	0.6907	0.4226	first simultaneous style and LPIPS gain
VQ + content-guided routing	1	0.6898	0.4156	stronger content preservation, style still limited
Prototype-aware latent target queue	1/3	0.6973	0.3331	major jump from better internal target distribution
Annealed prototype queue (F)	1	0.6969	0.3186	current LPIPS reference point
Hard-explore prototype queue (H)	2/1	0.6994	0.3213	current balanced point
Content-adaptive VQ atom routing (K)	1	0.7010	0.3623	style boost, but larger endpoint movement

Table 7: Destructive ablations under the historical strict-750 protocol.

Variant	CLIP-S	CLIP-C	LPIPS	EC
D0 full	0.7014	0.8022	0.4593	0.3791
D1 w/o SA-SWD	0.6708	0.8989	0.3490	0.4368
D2 w/o kinetic	0.7159	0.6624	0.6375	0.2596
D8 strong color loss	0.6923	0.6629	0.5675	0.2994

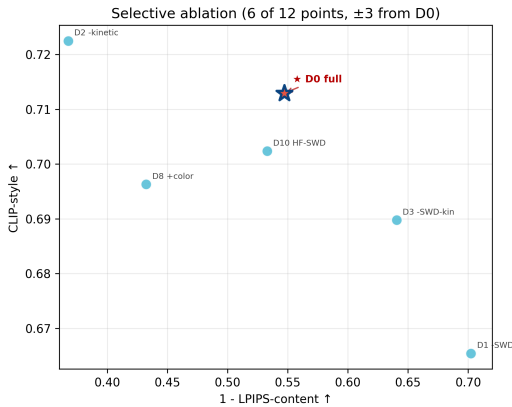


Figure 4: Destructive ablations. Removing SA-SWD improves content but weakens style, while removing kinetic regularization increases style at the cost of severe content degradation.

best raw-style result in the sweep was K1 balanced default at epoch 8 with CLIP-S 0.7161, CLIP-C 0.7984, LPIPS 0.4605, and EC 0.3863. The practical reading is simple: more complex category weighting was not consistently better than balanced sampling, so we keep this sweep as a supporting sensitivity result rather than spending main-figure budget on it.

Step-count sensitivity is also stable. Evaluating 1, 4, 8, 12, and 16 integration steps on the selected epoch-7 checkpoint changes all-pairs CLIP-S only from 0.7159 to 0.7162, while LPIPS remains near 0.4514. Four steps already recover nearly identical quality, with measured strict-750 throughput increasing from 8.43 images/s at one step to 9.54 images/s

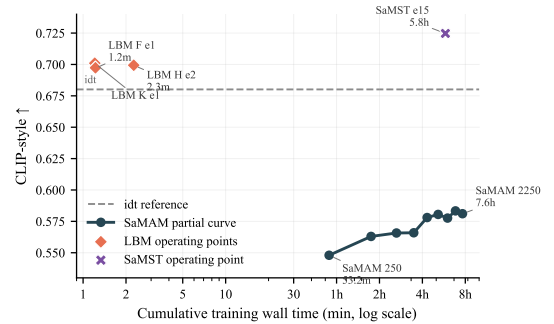


Figure 5: Same-scope Distinct5 timing-context artifact under the reproduced full 5×5 / 750-output protocol, with cumulative training wall time and evaluation excluded from the horizontal axis. LBM is shown as reviewed operating-point records, SaMAM as bounded pre-2250 timing records with the active manuscript boundary fixed at the trusted 2250 checkpoint, SaMST as the currently available operating point, and *idt* as the unchanged-image reference. This figure is used as bounded timing context rather than as a closed normalized time-to-parity claim.

at four steps and remaining around 9.34–9.50 images/s for 8–16 steps. A 3×3 grid over $\lambda_{\text{kin}} \in \{0.5, 1.0, 2.0\}$ and $\lambda_{\text{swd}} \in \{10, 20, 30\}$ produced a smooth style-content surface: higher kinetic weight and lighter terminal pressure improve content, while weaker kinetic weight or stronger terminal pressure increases style at the cost of LPIPS.

All reported evaluations use fixed seeds, stored resolved configurations, source snapshots, checkpoints, and per-epoch logs. Historical timings were measured on an RTX 4070 Laptop GPU unless otherwise noted. For Distinct5-512, Figure 5 consolidates the timing evidence into one explicit same-scope packet: cumulative training wall time is plotted with evaluation excluded, and each method family is labeled by its actual operating mode so the practical footprint can be compared without conflating unlike training protocols. For the historical table, training and inference cost are reported explicitly in Table 4 rather than being folded into a single efficiency slogan.

Discussion and Limitations

The main lesson is that style transfer quality cannot be inferred from raw CLIP-style alone. On the historical strict-750 protocol, StyleID has the highest raw style but collapses content, SaMST is slightly stronger than LBM on CLIP-S and CLIP-C but shows grain-like local artifacts, and LBM occupies a cleaner LPIPS/artifact-sensitive operating point. Distinct5-512 sharpens the same lesson further because its no-op baseline is measured explicitly: higher raw style is achievable, but only by moving into a severe LPIPS and targetwise ArtFID failure region, and some baselines fail even to exceed the identical-image prior. This exposes a community-relevant blind spot in art-to-art evaluation: preserved art-domain priors can masquerade as target-style success unless the unchanged-image floor is reported. Earlier AST evaluation work already notes that protocols are inconsistent across papers and that automatic scores can be style-dependent or misaligned with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our claim is therefore concrete: for separated art-to-art stress splits of this type, reporting the unchanged-image reference and Δ_{idt} is necessary to disentangle genuine style movement from preserved art-domain affinity. Across both protocols, the conclusion is consistent: on Distinct5-512 and within the currently reproduced positive- Δ_{idt} operating points, LBM gives the strongest measured CLIP-S / LPIPS trade-off, while the remaining open mechanism question is how faithfully style-side distinctions survive execution rather than whether code-space changes can be induced at all.

The tokenizer experiments narrow the design question. Within the tested Distinct5 design space, raw code capacity alone is not the bottleneck. The sharper issue is whether target-style information survives execution through a content-conditioned vector field strongly enough to yield real style gain beyond the unchanged-image reference. The current evidence therefore favors target-style carriers, prototype-aware target queues, and content-aware routing over undifferentiated token expansion. Matched successor probes reinforce the same point: tokenizer code separation maps only moderately into executed separation, while style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. This keeps executed style survival as the sharper mechanism question in the current successor family, and it suggests that the next representation stage should factor target representation from execution budget rather than simply adding parameters.

The study has several limitations. First, the main protocol is domain-level rather than reference-image-specific, so it does not attempt to reproduce a unique style image. Second, Euler integration is deliberately simple; although the step sweep is stable in the reported regime, more aggressive paths or higher resolutions may need better solvers or stronger constraints. Third, SA-SWD is an adaptive-projection distribution loss; current controls show that most of its benefit comes from the distributional term rather than from the specific semantic projection rule. Fourth, artifact-sensitive metrics are still proxies for visual preference; human studies or rubric-based VLM assessment would strengthen the claim further. Finally, Distinct5-512 is a five-domain stress split

rather than a universal art benchmark, so broader domain coverage remains future work even though the current cross-method comparison is already informative on this split.

Conclusion

We presented Latent Bridge Matching, an OT-coupled latent transport model for compact domain-level artistic transfer. The method separates endpoint construction, path-wise field learning, adaptive-projection terminal distribution matching, and style-tokenized execution. Under the reproduced historical strict-750 protocol, LBM delivers the cleanest reproduced balance of style strength, content fidelity, artifact control, and practical footprint within the mixed-family comparison packet considered here. The Distinct5-512 stress benchmark sharpens the paper’s main evaluation point: once the identical-image prior is made explicit, LBM stays above the no-op floor and defines the strongest measured positive- Δ_{idt} CLIP-S / LPIPS frontier among the reproduced compact operating points, while lower ArtFID alone does not guarantee useful target-style movement. For separated art-to-art stress splits of this type, reporting the IDT floor together with Δ_{idt} is therefore a necessary diagnostic rather than an optional extra statistic. The tokenizer study then isolates the remaining mechanism question. Within the tested Distinct5 tokenizer variants, matched localization controls show that executor-side refresh recovers more than style-side refresh alone on the current local surface, while the same-family H path-stability packet supports the bounded claim that kinetic regularization acts as a practical stabilizer in the current OMF / field regime. The next representation target is therefore to preserve target-style separation through execution itself, via explicit target carriers and tighter content-risk control.

References

- Bińkowski, M.; Sutherland, D. J.; Arbel, M.; and Gretton, A. 2018. Demystifying MMD GANs. In *International Conference on Learning Representations*.
- Botti, F.; Ergasti, A.; Rossi, L.; Fontanini, T.; Ferrari, C.; Bertozzi, M.; and Prati, A. 2025. Mamba-ST: State Space Model for Efficient Style Transfer. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 7786–7795.
- Chung, J.; Hyun, S.; and Heo, J.-P. 2024. Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Cuturi, M. 2013. Sinkhorn Distances: Lightspeed Computation of Optimal Transport. *Advances in Neural Information Processing Systems*.
- Deng, Y.; He, X.; Tang, F.; and Dong, W. 2024. Z*: Zero-shot Style Transfer via Attention Reweighting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 6934–6944.
- Deng, Y.; Tang, F.; Dong, W.; Sun, W.; Huang, F.; and Xu, C. 2022. StyTr2: Image Style Transfer with Transformers.

- In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Ding, K.; Ma, K.; Wang, S.; and Simoncelli, E. P. 2020. Image Quality Assessment: Unifying Structure and Texture Similarity. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2016. Image Style Transfer Using Convolutional Neural Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Gim, H.; Yoo, J.; and Uh, Y. 2024. Content-Adaptive Style Transfer: A Training-Free Approach with VQ Autoencoders. In *Proceedings of the Asian Conference on Computer Vision*.
- Heusel, M.; Ramsauer, H.; Unterthiner, T.; Nessler, B.; and Hochreiter, S. 2017. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. In *Advances in Neural Information Processing Systems*.
- Hong, K.; Jeon, S.; Yang, H.; Fu, J.; and Byun, H. 2023. AesPA-Net: Aesthetic Pattern-Aware Style Transfer Networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Huang, X.; and Belongie, S. 2017. Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization. In *Proceedings of the IEEE International Conference on Computer Vision*.
- Huang, X.; Zhang, Z.; Zhang, J.; et al. 2024. AesFA: An Aesthetic Feature-Aware Arbitrary Neural Style Transfer. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Jiang, Y.; Jiang, L.; Yang, S.; Liu, J.-W.; Tsang, I. W.; and Shou, M. Z. 2025. Balanced Image Stylization with Style Matching Score. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 17346–17355.
- Ke, J.; Wang, Q.; Wang, Y.; Milanfar, P.; and Yang, F. 2021. MUSIQ: Multi-scale Image Quality Transformer. *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Léonard, C. 2014. A Survey of the Schrödinger Problem and Some of Its Connections with Optimal Transport. *Discrete and Continuous Dynamical Systems - A*.
- Liu, H.; Wang, L.; Zhang, Y.; Yu, Z.; and Guo, Y. 2025. SaMam: Style-aware State Space Model for Arbitrary Image Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 28468–28478.
- Liu, Y.; et al. 2024. Pluggable Style Representation Learning for Multi-Style Transfer. In *Proceedings of the Asian Conference on Computer Vision*.
- Park, D. Y.; and Lee, K. H. 2019. Arbitrary Style Transfer with Style-Attentional Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Peyré, G.; and Cuturi, M. 2019. Computational Optimal Transport. *Foundations and Trends in Machine Learning*.
- Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, 8748–8763.
- Shang, C.; Wang, Z.; Wang, H.; and Meng, X. 2025. SCSA: A Plug-and-Play Semantic Continuous-Sparse Attention for Arbitrary Semantic Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 13051–13060.
- Wright, M.; and Ommer, B. 2022. ArtFID: Quantitative Evaluation of Neural Style Transfer. *arXiv preprint arXiv:2207.12280*.
- Xu, R.; Xi, W.; Wang, X.; Mao, Y.; and Cheng, Z. 2025. StyleSSP: Sampling StartPoint Enhancement for Training-free Diffusion-based Method for Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 18260–18269.
- Yang, S.; Wu, T.; Shi, S.; Lao, S.; Gong, Y.; Cao, M.; Wang, J.; and Yang, Y. 2022. MANIQA: Multi-dimension Attention Network for No-Reference Image Quality Assessment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*.
- Yeh, M.-C.; Tang, S.; Bhattad, A.; Zou, C.; and Forsyth, D. 2020. Improving Style Transfer with Calibrated Metrics. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 3160–3168.
- Zhang, R.; Isola, P.; Efros, A. A.; Shechtman, E.; and Wang, O. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Zhang, S.; Kang, H.; Liu, Y.; Mei, F.; and Li, H. 2025. HSI: A Holistic Style Injector for Arbitrary Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 23433–23442.
- Zheng, S.; Gao, P.; Zhou, P.; and Qin, J. 2024. Puff-Net: Efficient Style Transfer with Pure Content and Style Feature Fusion Network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 8059–8068.
- Zhou, X.; Zeng, T.; Xu, H.; Liu, X.; Zhao, J.; and Yao, Y. 2024. A Comprehensive Evaluation of Arbitrary Image Style Transfer Methods. *IEEE Transactions on Visualization and Computer Graphics*, 30(1): 1–19.

Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides three formal results with proofs

and partial empirical support: a path-action decomposition connecting the training kinetic loss to inference-time path energy (Theorem 1), an $O(1/K)$ Euler discretization error bound (Theorem 2), and a Lipschitz-continuity result for the OT assignment map explaining its directional-coherence benefit (Theorem 3). Full proofs are in the supplement. These results justify the three-part design without claiming equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **No**.
- All source code required for conducting and analyzing the experiments is included in a code appendix: **No**.
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial**.
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Partial**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **Partial** (selected confidence intervals and bootstrap summaries are reported, but not every manuscript comparison is presented as a closed significance claim).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.